Certifying the Outcome of Autonomous Vehicle Fallback Triggers: Closed-Loop Evaluation Under Latency and Dataset Shift

Hassan Waqas Saleem and Rahat Zeeshan Saleem

***Abstract*—A fallback trigger decides when an automated vehicle should abandon its task for a minimum-risk maneuver. Statistical certificates for such triggers bound the alert, not what happens once the maneuver executes. Under determinism assumptions that we measure, a rollout depends only on the tick at which the fallback fires, so one reference run plus one forced rollout per decision tick evaluates every trigger, threshold and latency by lookup, which agreed with direct execution in all 1,008 checked pairs. On replayed Argoverse 2, nuScenes and Waymo logs with reactive traffic, a Learn-then-Test certificate keeps its 5 % miss target (3.7 % missed). No resplit clearly violates it, unlike 12–15 % of resplits for tuned, risk-control and confidence thresholds. The alert certificate still leaves 6.5 % harm after the fallback, so we certify the executed outcome with risk-ordered fixed-sequence testing. At a 7.5 % target the realized harm is 5.4 %. A certifiability map shows that a time-to-collision threshold is the cheapest at that target up to 0.5 s of latency (6–13 % unnecessary stops at 10 Hz) and meets it at zero latency on six of eight shifted test sets. Targets 1–2 points stricter need geometric or learned triggers, and delays of 0.8 s or more need learned or safe-distance triggers, at 30–50 % stops. Validity replicates with three predictors, two traffic models and fresh data. Certificates fail on nuScenes and in two of six held-out cities, the targets with the highest harm rates, and hold on Waymo.**

***Index Terms*—Autonomous driving, conformal prediction, minimum-risk maneuver, distribution shift.**

I. Introduction

AUTONOMOUS vehicles that lack confidence in their own situational understanding are expected to fall back on a minimum-risk maneuver (MRM): slow to a stop, pull to the shoulder, or hand control back to a human. When to invoke it is, in most deployed and published systems, a hand-tuned threshold on a scalar confidence, uncertainty or risk score from a perception or prediction module. Such a threshold carries no guarantee, and learned uncertainty estimates are known to degrade under the shift a vehicle meets in a new city or weather. The failure is silent: the system keeps reporting a confidence number that is no longer right.

This work received no external funding.

H. W. Saleem (corresponding author) is with a public-sector organization in Pakistan; he received the M.S. degree from the School of Electrical Engineering and Computer Science (SEECS), National University of Sciences and Technology (NUST), H-12, Islamabad 44000, Pakistan (e-mail: hsaleem.msee20seecs@seecs.edu.pk; ORCID: 0009-0003-4461-8005).

R. Z. Saleem is with a private-sector organization in Pakistan; he received the M.S. degree from the NUST Business School (NBS), National University of Sciences and Technology (NUST), H-12, Islamabad 44000, Pakistan (e-mail: rahat.mscm21nbs@student.nust.edu.pk).

Conformal prediction offers a principled alternative, a distribution-free, finite-sample guarantee under exchangeability, and has been applied widely to trajectory forecasting and planning in driving. A guarantee on a prediction, or on a warning, is nevertheless not a guarantee on the consequences of the action it triggers. Whether an alert leads to a fallback is rarely evaluated in closed loop, and triggering is not free: a vehicle that stops in a live lane because its uncertainty grew is not obviously safer than one that continues, and a trigger that fires whenever asked trivially achieves a good miss rate while being useless.

We ask a narrower question than "which trigger is best": *can the outcome of a fallback trigger, not only its alert, be certified, and how does that certificate behave under actuation latency and dataset shift?* Certifying a threshold on a risk score against missed events is not new: Luo *et al.* [27] do so with conformal prediction, and Conformal Decision Theory [31] calibrates decisions through a long-run *average* guarantee. We do not claim the certification step. Three elements are new, to our knowledge. First, the guarantee is placed on the harm in the run the vehicle actually drives, not on the alert. Second, timing is part of the problem: a fallback is sequential and absorbing, so *when* it fires determines every later outcome, unlike a per-instance switch to a fallback policy [18] or an abort alert evaluated without its response [46]. Third, a lookup protocol makes this evaluation exhaustive and cheap. A deployed vehicle needs such a guarantee before the first mile, and a minimum-risk maneuver is irreversible and its errors are paid in two currencies, missed interventions and the harm of stopping itself.

We organize the study around four research questions, each tied to a hypothesis fixed before data collection (Section IV-G). **RQ1 (H2):** does a certificate keep its promised miss rate in closed loop, and what does it cost in unnecessary stops and induced collisions? **RQ2 (H1):** do open-loop metrics choose the trigger that closed-loop safety does? **RQ3:** how sensitive is a trigger to latency, and can the outcome be certified under it? **RQ4 (H3, H3-W):** what happens to a certificate under cross-dataset shift, and how much can unlabeled or a few labeled target scenarios repair?

We make four contributions.

1) **A lookup evaluation protocol for executed fallbacks.** Under a determinism assumption whose violation we measure, a rollout's outcome depends only on *when* the fallback fires (Observation 1, Fig. 1). One reference run plus one rollout forced to fire at each decision tick (about

17 per scenario) evaluates every trigger, threshold, online method and latency on identical physics, where direct execution would need over 7,000 rollouts per scenario here. We check the lookup against direct execution (Section III-E).

2) **Certificates that keep their promise, on the alert and on the outcome.** A Learn-then-Test certified alert keeps its miss target where tuned, risk-control and confidence thresholds are clearly violated, and a noise-corrected violation measure separates the two. Because the alert certificate leaves 6.5 % harm after the fallback, risk-ordered fixed-sequence testing certifies the harm in the run the vehicle actually drives, a non-monotone loss. At a 7.5 % target the certified threshold realizes 5.4 %.
3) **A certifiability map, with a certified physics trigger as the default.** We map which triggers can be certified at which latency and target. A certified time-to-collision (TTC) threshold is the cheapest at a 7.5 % target up to 0.5 s of latency at 10 Hz (6–13 % unnecessary stops against 19–33 % for safe-distance, geometric and learned triggers) and, at zero latency, meets that target on six of eight shifted test sets. Targets 1–2 points stricter need geometric or learned triggers, and delays of 0.8 s or more need learned or safe-distance triggers, at 30–50 % stops.
4) **A failure map for the certificate.** An Argoverse 2 certificate fails on nuScenes and in two of six held-out cities but holds on Waymo, with failures tracking the target's harm rate, and spatial dependence weakens the high-probability statement, which a cluster-aware certificate restores at a cost.

Section V-I reports, in one place, the pre-registered claims that did not hold, and analyses run after the pre-registered verdicts are labeled post hoc.

II. Related Work

A. Conformal prediction for autonomous driving

Conformal prediction [1], [9] gives prediction sets a distribution-free, finite-sample coverage guarantee under exchangeability and has been applied widely to trajectory forecasting and safe planning [10]: per scenario type [11], with feedback from realized trajectories [12], with CVaR clearance margins [13], and in crowd navigation [14]. Lindemann *et al.* [4] embed conformal regions in a model predictive controller, a guarantee that collapses under shift, which Rahaman *et al.* [23] address with robust conformal prediction; SafePath [24] guarantees that a set of LLM-planned paths contains a safe one. In these works the guarantee is on the *prediction* or a candidate set, not on a fallback-triggering decision replayed on driving logs under reactive traffic. Ours is on the decision, and in Section III-F on its executed outcome.

B. Online and decision-level conformal methods

Adaptive Conformal Inference (ACI) [32] updates the miscoverage level online, and Dixit *et al.* [5] use it inside a model predictive controller to size a prediction set, a guarantee on the region, not on a fallback decision. Conformal Decision Theory (CDT) [31] carries the idea to decisions: it updates its threshold online so that the long-run average risk converges to ε. Both give *asymptotic, average* guarantees. We target a one-shot, irreversible trigger whose certificate must hold with finite samples before deployment, and adopt both as baselines (Section IV-E). Offline risk control now covers non-monotone losses [15], several risks at once [16], and planning with control barrier functions [17]. Closest to our accounting of what the fallback does, Joshi *et al.* [18] certify the outcome of switching to a fallback policy, per instance and with no notion of timing. An MRM trigger is sequential and absorbing: *when* it fires determines every outcome after it.

C. Task-relevant and closed-loop failure detection

Farid *et al.* [25] flag a misprediction only when the resulting planning cost is anomalous, evaluated without an actuated fallback. Our labeling reflects this insight: a collision suffered while stopped is a cost of the fallback, never a missed intervention.

Closest to our certification step, Luo *et al.* [27] tune the threshold of a warning system with Mondrian conformal prediction so that, among situations that are unsafe in the true future *in the absence of corrective action*, fewer than ϵ raise no alert. Their alerts are scored against logged data, so the alert triggers nothing. We adopt the same principle and do not claim it as new. The differences are the guarantee (class-conditional there, marginal over scenarios here), the consequence of an alert (none there, an executed maneuver here), and dataset shift and latency, which they do not study (Supplementary Table S1). We implement their certificate as a baseline (Supplementary Section S1).

Sinha *et al.* [38] use a conformal runtime monitor to keep a fallback plan of a robust MPC controller feasible, so their fallback is part of the controller design, whereas ours is an irreversible maneuver whose trigger is the object of study. Atoui and Resende [19] design and validate an MRM controller, taking the decision to engage it as given. Scenario-based simulation is a standard route to evaluating automated vehicle safety [39], [40] that needs a statistical foundation [20] and can use risk-based filtering [21]. We replay real logs with reactive traffic.

Fallback triggering also sits within established safety frameworks. Runtime monitors that switch to a trusted fallback, as in [19], [38], follow the runtime-assurance idea of a monitored learned component with a trusted backup, as in the Simplex architecture [49]. Switching policies have been learned for this purpose, with safety from reachability [45], but without a statistical certificate on the executed outcome, and conformal abort alerts have been calibrated in flight testing [46] without evaluating the response. Responsibility-Sensitive Safety [41] gives model-based rules for when a vehicle is at fault, whereas our trigger is certified from data. SOTIF [42] and UL 4600 [43] frame assurance at the system level, but do not specify how to certify a trigger threshold. Selective prediction [44] has the same reject-option structure, except that abstention here is an executed maneuver with its own cost. We do not claim priority in any of these areas. Conditional Value-at-Risk [28] would extend our analysis to the severity of violations, which we leave to future work.

III. Problem Formulation and Method

A. Fallback triggering as selective prediction

Consider a closed-loop rollout of a scenario X under a fallback policy governed by a scalar threshold λ: at every decision tick the vehicle computes a trigger score $u_t(X)$, mainly from a trajectory predictor's output for nearby agents (physics triggers use time-to-collision or distance), and engages the minimum-risk maneuver the first time $u_t(X) > \lambda$, remaining engaged for the rest of the episode. Define the closed-loop loss

$$L(X, \lambda) = \mathbb{1}\big[\text{harm in } X \text{ under } \lambda,\ \text{MRM not first}\big], \quad (1)$$

the *missed-intervention* indicator, and the complementary over-conservatism quantity

$$S(X, \lambda) = \mathbb{1}\big[\text{MRM fires under } \lambda,\ X \text{ harm-free at } \lambda = \infty\big], \quad (2)$$

the *unnecessary stop* indicator, evaluated against the counterfactual rollout of the same scenario with the trigger disabled. The goal is to choose $\hat{\lambda}$ such that

$$\Pr_{X_1,\ldots,X_n}\Big(\mathbb{E}_X\big[L(X, \hat{\lambda})\big] \le \alpha\Big) \ge 1 - \delta, \quad (3)$$

while minimizing $\mathbb{E}_X[S(X, \hat{\lambda})]$, the smallest autonomy loss compatible with the risk target α.

Let $h(X)$ be the first harm step of the reference run with the fallback disabled (∞ if none) and $\tau_\lambda(X)$ the firing step. Because a run under the trigger coincides with the reference run until it fires (Section III-E), $L(X, \lambda) = \mathbb{1}[h(X) \le \tau_\lambda(X)]$ and $S(X, \lambda) = \mathbb{1}[\tau_\lambda(X) < \infty,\ h(X) = \infty]$: both alert-level indicators depend only on the reference run and the firing step. The consequences of the fallback after it fires enter through separate quantities, which we report but do not certify: residual harm (harm anywhere in the executed run), induced collisions, and route completion. Residual harm is not monotone in λ, since firing earlier can itself create harm, and it includes harm after the alert, so (3) does not bound it. Section III-F certifies it directly. Since the running maximum of the score is non-decreasing, τ_λ is non-decreasing in λ and $L(X, \lambda)$ is non-decreasing in λ for every scenario. Equation (3) is therefore a monotone-risk statement, to which Conformal Risk Control [3] applies directly, but with an expectation guarantee. It bounds the calibration-averaged risk, not the probability that a given calibrated threshold exceeds α. We use Learn-then-Test [2] for the high-probability form. For a finite grid $\lambda_1 > \lambda_2 > \cdots > \lambda_J$ ordered from least to most conservative, we compute the empirical risk $\hat{R}(\lambda_j) = \frac{1}{n}\sum_i L(X_i, \lambda_j)$ on n calibration scenarios and a Hoeffding–Bentkus p-value for the null hypothesis $H_j : \mathbb{E}[L(\cdot, \lambda_j)] > \alpha$, then apply fixed-sequence testing starting from the most conservative λ. Every λ_j whose null is rejected satisfies (3) simultaneously, with family-wise error controlled at δ, and $\hat{\lambda}$ is the least conservative such λ_j (largest autonomy). The guarantee is therefore about the miss rate of the alert, over scenarios exchangeable with the calibration scenarios, with confidence $1 - \delta$ over the calibration draw. We report the violation frequency of CRC and LTT over resplits rather than assuming it.

The marginal target is weaker than it may sound. With harm in a fraction π of scenarios, a marginal miss probability α leaves up to a fraction α/π of harmful scenarios unalerted, which is $32\,\%$ at $\alpha = 0.05$ and $\pi = 0.158$ in our data ($24\,\%$ observed, Section V-F).

B. Trigger score

At each decision tick the geometric score u_t measures how far a predicted future position of any nearby agent intrudes into the ego vehicle's own near-term extent. The ego's constant-heading, constant-speed extrapolation over T steps, $\{p_\tau^{\text{ego}}\}_{\tau=1}^{T}$, is the same route-agnostic motion model used by the harm metric (Section IV-B), deliberately not the planned path. Each predicted offset of agent i, mode k, is split into a longitudinal component $a_\tau^{(i,k)}$ and a lateral component $\ell_\tau^{(i,k)}$ with margins $m_i^{\text{lon}} = \frac{1}{2}(L_{\text{ego}} + L_i) + c_{\text{lon}}$ and $m_i^{\text{lat}} = \frac{1}{2}(W_{\text{ego}} + W_i) + c_{\text{lat}}$. The intrusion is

$$\rho_\tau^{(i,k)} = \max\left(0,\ 1 - \sqrt{\left(\frac{a_\tau^{(i,k)}}{m_i^{\text{lon}}}\right)^2 + \left(\frac{\ell_\tau^{(i,k)}}{m_i^{\text{lat}}}\right)^2}\right) m_i^{\text{lat}}, \quad (4)$$

and $u_t = \max_{i,k,\tau} \rho_\tau^{(i,k)}$. The length-aware box is a modeling choice made during development. Our ablation (Supplementary Section S14) finds no measurable benefit over a width-only isotropic radius, so we report it as a design option.

C. Fallback maneuver

On triggering, the vehicle executes the minimum-risk maneuver defined by UN Regulation No. 157 (Automated Lane Keeping Systems) § 5.5.1 [37]: decelerate inside the current lane with a deceleration demand not exceeding $4.0\,\mathrm{m/s^2}$, lane-keeping control otherwise unchanged, holding the vehicle stationary once stopped. This is enforced by a proportional controller tracking the target speed profile $v(t) = \max(v_0 - 4.0\,t,\ 0)$ from the moment of triggering.

D. Weighted and cluster-aware repairs (heuristics)

The certificate in (3) assumes calibration and deployment scenarios are exchangeable. For covariate shift we reweight calibration scenarios by an estimated density ratio $w(x)$ from a domain classifier on pre-deployment covariates (agent counts and speeds, initial gap, route length) and unlabeled target scenarios, following weighted conformal prediction [33], and test the weighted risk with the Hoeffding–Bentkus p-value at the Kish effective sample size $n_{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2$. Because the weights are estimated and clipped and n_{eff} is an approximation, this is a heuristic repair with no finite-sample guarantee. As the weights concentrate, n_{eff} shrinks and the test certifies a more conservative threshold, or none. Spatial dependence is handled the same way: evaluating the p-value at the cluster effective sample size $n_{\text{eff}} = N^2 / \sum_g n_g^2$ of the spatial groups in the calibration set gives a cluster-aware certificate, also a heuristic.

E. Evaluating every trigger by lookup

Fix a scenario X and its random seed. The *reference run* $R(X)$ disables the fallback. For a decision tick j, the *forced*

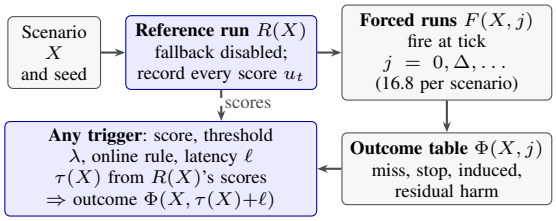


Fig. 1. Lookup evaluation (Observation 1). One reference run and one forced rollout per decision tick are simulated once; every trigger, threshold, online rule and latency is then evaluated by lookup (over 7,000 rollouts per scenario otherwise).

run $F(X, j)$ is identical to $R(X)$ until tick j and then executes the MRM for the rest of the episode. Let $\Phi(X, j)$ denote any outcome of $F(X, j)$ (miss, stop, induced collision, residual harm, route completion) and $\Phi(X, \infty)$ the outcome of $R(X)$. Fig. 1 summarizes the procedure.

Observation 1 (Firing-tick sufficiency): Assume (A1) the simulator, background traffic and predictor are deterministic functions of the seed and of the ego's trajectory, so two runs with the same seed and the same ego control up to a step have identical states up to that step; (A2) the trigger's decision at tick j depends only on the state history up to tick j; (A3) once the fallback fires it is absorbing, and the control after firing depends only on the state at firing. Then for every trigger with firing tick $\tau(X) \in \{0, \Delta, 2\Delta, \ldots\} \cup \{\infty\}$, the run under the trigger is $F(X, \tau(X))$, and every outcome equals $\Phi(X, \tau(X))$. This holds for any score, threshold, online update rule and actuation latency ℓ (for which the run is $F(X, \tau(X) + \ell)$).

Proof: Until τ the control is that of $R(X)$ by (A3) and the states equal those of $R(X)$ by (A1), so by (A2) the trigger's decisions, and hence τ, are determined by $R(X)$ alone. At τ the control depends only on the state at τ (A3), as in $F(X, \tau)$, and both runs have equal states afterwards by (A1). ∎

Alert-level indicators need no forced run, since L and S are functions of $R(X)$ and τ alone, and the consequences of the fallback need only 16.8 forced rollouts per scenario on average (at most 26), where direct execution needs over 7,000. The observation is elementary and its value is practical. Assumption (A1) is an idealization, so we measure its violation. Of the first 33,761 forced rollouts, 98.4 % agreed with the reference run to within 1 cm before firing and one changed the pre-fire harm status (Section V-D drops the 3.2 % of scenarios with any flagged drift as a check). When four triggers were executed directly on 252 random test scenarios, firing step, miss, unnecessary stop and induced collision agreed with the lookup in all 1,008 pairs (upper bound 0.4 %), and harm anywhere in the run in all but one (Supplementary Section S7).

F. Outcome-level certification and learned triggers

The forced rollouts also give, for every scenario, the outcome of firing at each tick. Let the *safe window* $W(X)$ be the set of firing ticks (with "never") whose executed run has no harm, $F(X)$ its first tick, and $O(X, \lambda) = \mathbb{1}[\text{harm in the run under } \lambda]$ the *outcome loss*, with a total-harm variant that also counts induced collisions. O is not monotone in λ and $W(X)$ is empty when harm is unavoidable. Learn-then-Test needs a monotone order only to be powerful, not to be valid: any order fixed before looking at the calibration data yields a valid family-wise guarantee. We therefore order the threshold grid by the risk of O estimated out of fold on a training split, run fixed-sequence testing along that order on a separate calibration split, and take among the certified thresholds the one with the fewest unnecessary stops, which is valid because all certified hypotheses hold simultaneously. A threshold chosen on one split also yields a one-sided Clopper–Pearson upper bound on outcome harm from a fresh split.

We also learn triggers. A gradient-boosted classifier predicts, at each decision tick, from the stored scores, time-to-collision, headway distance, agent counts, predicted gaps and scenario covariates, whether firing is warranted, and its predicted probability is the trigger score. We compare two labels: a *counterfactual* label (the no-fallback run is harmful and $t \geq F(X)$, so firing is neither unnecessary nor too early) and a *factual* label (harm occurs within 2 s after the action takes effect). For a delay of ℓ the labels are shifted by ℓ, because a decision at t acts at $t + \ell$. All hypotheses on these triggers were pre-registered (Section IV-G).

IV. Experimental Setup

A. Simulation environment

All rollouts are executed headless in MetaDrive, replaying scenarios converted by ScenarioNet [29] from Argoverse 2 Motion Forecasting [6], nuScenes [7] and Waymo Open Motion [26]. Non-ego traffic is controlled reactively by the Intelligent Driver Model [8], not replayed open-loop from the log. Pure log replay roughly tripled the no-fallback collision rate on a 1,000-scenario diagnostic sample, because logged agents cannot react to an ego vehicle that deviates from its log, precisely what a fallback maneuver produces, and would misrepresent the cost of stopping. The ego follows its logged route under IDM longitudinal control (MetaDrive's TrajectoryIDMPolicy) until the fallback engages, so the study concerns the trigger, not planning quality.

B. Harm labeling

A rollout is labeled harmful using the nuPlan closed-loop safety definition [30]: an at-fault collision, or a time-to-collision below 0.95 s computed over a 3.0 s horizon at 0.1 s resolution, not evaluated while the ego is stationary or against tracks behind it. Fault attribution follows the nuPlan convention. A collision the ego suffers while stationary, or while struck from behind, is not at-fault and never counts as a missed intervention. It is recorded as an *induced collision*, the measurable cost of a false or premature trigger.

C. Predictor

The trajectory predictor is AutoBot [35], one of many learned multi-modal forecasters [22] (6 modes, 2.1 s history, 6 s horizon), trained with the UniTraj framework [36] on the Argoverse 2 Motion Forecasting training split (minADE$_6$ = 1.09 m on held-out validation scenes). It runs online inside the closed loop, so its input is the actual, possibly ego-deviated,

simulator history, with the UniTraj preprocessing unchanged. Vehicles are predicted by the network. Pedestrians and cyclists, which the checkpoint was not trained on, receive constant-velocity forecasts. The robustness checks add a second AutoBot checkpoint (minADE$_6$ = 0.854) and Wayformer [47].

D. Datasets and splits

All scenarios come from Argoverse 2 validation, disjoint from the predictor's training data, split deterministically by scenario identifier into development (14,990), calibration (4,971) and test (5,027) sets (design choices used the development split only). The shifted targets are nuScenes (9,041 scenarios of 8.1 s, Boston and Singapore) and, under a separate pre-registered amendment, 1,000 Waymo Open Motion [26] validation scenarios (9.1 s). Scenarios in which the ego moves less than 10 m are excluded, as in ScenarioNet. The pre-registered evaluation uses 2,000 calibration, 2,000 test and 1,500 nuScenes scenarios, of which ten Argoverse 2 scenarios with unmatched per-agent records are dropped from every analysis, leaving 1,994, 1,996 and 1,504. Decisions are taken at 2 Hz, with identical miss rates at 10 Hz on a paired development comparison. The latency map of Section V-D uses a separate 10 Hz arm, because a 2 Hz tick would make every delay a multiple of 0.5 s.

E. Baselines

All methods use the same predictor, the same MRM and the same test scenarios, and are evaluated by lookup (Section III-E).

- **T0 — time-to-collision threshold** (physics trigger, no learned component): fire when the nuPlan time-to-collision, projected at constant speed and heading, falls below a threshold; certified with LTT or tuned. Headway distance and an RSS-style safe-distance trigger [41] (response time 0.5 s, accelerations 3, braking 4 and 8 m/s^2, fixed beforehand) are two more physics variants.
- **T1 — confidence threshold** (uncertainty practice): fire when $1-\max_k p_k$ of the network-predicted vehicle whose predicted modes come closest to the ego exceeds a threshold. Agents the network does not predict (pedestrians, cyclists, distant vehicles) are excluded from this score. The threshold is tuned on calibration data to the most permissive value with empirical miss rate $\leq \alpha$. A variant that uses the nearest agent ahead is also evaluated.
- **T2 — ensemble threshold**: the same tuning, applied to the spread of endpoint predictions across five Monte Carlo dropout passes [34].
- **T3 — open-loop conformal**: inflate every predicted mode by the split-conformal radius of the predictor's 3 s error on held-out Argoverse 2 logs at level α [1], and fire when the inflated region touches the ego.
- **Tuned geometric**: our score with an empirically tuned threshold and no certificate. It isolates the contribution of the calibrator.
- **CDT** [31] and **ACI** [32]: online decision- and prediction-level adaptation, updated after every scenario. Each uses the step size that performs best on test, which favors the baselines. Choosing it on calibration data instead gives CDT 4.6 % missed and 25.1 % stops, and ACI the same result.
- **Oracle**: the best threshold in hindsight on test data. It is not deployable and is reported as an upper bound.

F. Metrics and statistics

For every method we report the missed-intervention, unnecessary-stop and fallback-induced-collision rates, autonomy (the fraction of scenarios completed without the fallback) and route completion, with Wilson 95 % intervals. Validity of each certificate is assessed over 200 random half-and-half resplits, without replacement, of the pooled 3,990 calibration and test scenarios. A valid certificate keeps the frequency of test miss rate above α at or below δ. The resplits overlap, so they are not independent draws: the frequency estimates the violation probability for this pool, with Monte Carlo error (Wilson 95 % interval [6.2, 14.4] for 9.5 %, [36.8, 50.4] for 43.5 %). Scenarios from one place may be dependent: resplitting whole spatial groups (ego starts within 150 m, a proxy) raises LTT's violation frequency to 17.0 % [12.4, 22.8], above δ (Supplementary Section S12). The cluster-aware certificate of Section III-D removes these violations for 50 m groups (0 % against 14 %) at 10 points more unnecessary stops, and refuses at 150 m, where one group holds 16 % of the pool and the effective sample size falls to about 10 scenarios (Supplementary Section S23). Harm uses the nuPlan definition above, and Section V-H repeats the key comparison under three alternatives.

G. Pre-registration and reproducibility

The harm definition, $\alpha = 0.05$, $\delta = 0.10$, and the refutation criteria for three hypotheses were recorded in a version-controlled note before any calibration or test data were collected: **H1**, that open-loop metrics rank triggers differently from closed-loop outcomes; **H2**, that a certified trigger achieves a better safety–conservatism trade-off than a tuned fixed threshold; and **H3**, that a threshold certified on Argoverse 2 loses its guarantee on nuScenes. A later amendment, written before any Waymo outcome existed, added **H3-W**, the same hypothesis on Waymo, and a few-label recalibration study. Further dated amendments were each written before the corresponding outcomes were evaluated, and analyses run afterwards are labeled post hoc. Five of them (Amendments 11 to 15: the TTC baseline, noise-corrected validity, outcome-level certification, the learned counterfactual trigger, the certifiability map, dependence checks and a cluster-aware certificate) were written after the main results were known, and twelve of their twenty-one pre-registered claims were refuted (Supplementary Section S17).

Data and code availability. Code, the pre-registration note with all amendments and per-scenario rollout records are available at https://github.com/hwaqas160/paper03-risk-fallback, and a snapshot is archived at https://doi.org/10.5281/zenodo.23054382. Each scenario's record contains every score trace and outcome, so every table can be regenerated without re-simulation.

TABLE I
CLOSED-LOOP OUTCOMES ON THE ARGOVERSE 2 TEST SET AT $\alpha = 0.05$. RATES IN %, UNNECESSARY-STOP RATE WITH ITS 95 % WILSON CI WHERE AVAILABLE. LAST COLUMN: FRACTION OF 200 CALIBRATION/TEST RESPLITS IN WHICH THE TEST MISS RATE EXCEEDS α (U, UNCORRECTED), AND IN WHICH ITS ONE-SIDED 95 % LOWER CONFIDENCE BOUND DOES (C, CORRECTED). VALID IF U $\leq$ 10. CDT AND ACI: 100 RESPLITS.

Method	Miss	Unnec.	95 % CI	Induced	Auton.	Viol. u/c
No fallback	15.1	0.0	[0.0, 0.2]	0.0	100	–
Always fire	0.0	84.9	[83.2, 86.4]	6.5	0	–
T0 TTC, LTT (certified)	3.1	7.1	[6.1, 8.3]	1.0	78.3	1.0 / 1.0
T0 TTC, tuned	3.5	5.7	[4.7, 6.8]	0.9	79.8	50.5 / 1.0
T1 confidence	4.6	68.3	[66.2, 70.3]	4.0	20.3	50.5 / 14.0
T2 MC-dropout spread	4.0	70.8	[68.8, 72.7]	4.9	17.6	–
T3 open-loop conformal	0.0	73.8	[71.8, 75.7]	5.5	11.1	0.0 / 0.0
Tuned geometric (no cert.)	4.3	26.0	[24.1, 28.0]	2.3	61.4	43.5 / 15.0
CRC (geometric)	4.3	26.0	[24.1, 28.0]	2.3	61.4	41.0 / 12.0
LTT (geometric, certified)	3.7	28.5	[26.5, 30.5]	2.6	58.5	**9.5 / 0.0**
CDT (online)	4.9	24.1	–	2.2	63.9	59.0 / 0.0
ACI (online)	1.3	45.0	–	3.5	40.2	0.0 / 0.0
Oracle (hindsight)	5.0	23.6	–	2.2	64.2	–

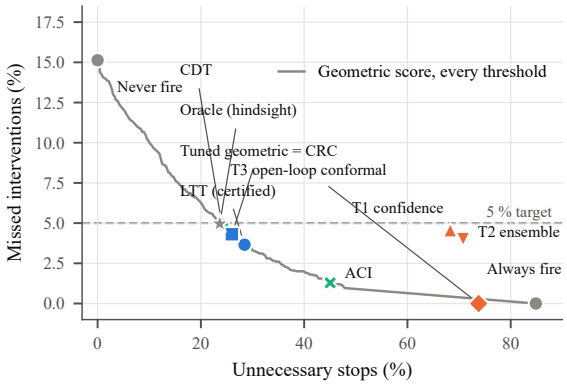


Fig. 2. Every method at its operating point (Table I), against the geometric score's own miss-versus-stops curve as its threshold varies. On this split most methods sit below the dashed $\alpha = 0.05$ line; the geometric methods do so at well under half the stops of the confidence, ensemble and open-loop triggers. Which of them keeps the target across resplits is in Table I.

V. RESULTS

All results use $\alpha = 0.05$ and $\delta = 0.10$, 1,994 calibration and 1,996 test scenarios from Argoverse 2, and 1,504 nuScenes scenarios, with Wilson 95 % intervals unless stated. Sections V-A–V-C, the shift study and the Waymo recalibration are pre-registered. Section V-D and the TTC baseline come from later amendments, and the remaining analyses are post hoc.

A. Does a certificate keep its promise, and what does it cost? (RQ1)

Table I gives every method at its operating point on the Argoverse 2 test set. The certified geometric trigger (LTT) fails to pre-empt harm in 3.7 % of all scenarios [95 % CI 2.9–4.6], below the 5 % target (24 % of the harmful scenarios), with 28.5 % unnecessary stops and 2.6 % fallback-induced collisions. The last column repeats the calibration on 200 random resplits. A valid certificate keeps the uncorrected frequency at or below $\delta = 10\,\%$, and LTT does (9.5 %). That count also includes test-sample noise, so we add a corrected count that only counts resplits whose test miss is clearly above α. Empirical tuning, Conformal Risk Control (CRC), which guarantees only an expectation, and the tuned confidence trigger are clearly violated under both counts (43.5 %, 41.0 % and 50.5 % uncorrected; 12–15 % corrected), while LTT is not violated under either. Among the predictor-based triggers, certification is therefore the only design that reliably keeps its target at a useful operating point. The uncertified TTC threshold, a physics trigger, sits far below the target and is only indeterminate. Fig. 2 plots every method against the geometric score's own curve as its threshold varies, and the resplit distributions behind the violation column are in Supplementary Fig. S1. CRC and tuning select the same threshold (no grid point lies between the calibration risks 4.81 % and 5.02 %).

A certified physics trigger is the best valid design. A time-to-collision threshold, certified with the same procedure, meets the target with 3.1 % missed, only 7.1 % unnecessary stops and 1.0 % induced collisions, and is not violated under either count (Table I), against 28.5 % stops for the predictor-based geometric trigger. Tuned without a certificate, the same TTC score exceeds the target in half the resplits, though within test noise. The comparison is not neutral, because the harm label is itself defined by time-to-collision, and it holds under three of the four harm definitions we tried, including collision only, but not under TTC < 1.5 s (Supplementary Section S14). Under this label the predictor adds nothing at zero latency, and the confidence and ensemble triggers, which stop 40–45 points more often than the geometric trigger, are weak comparators, in part because this predictor's mode probabilities are weakly informative (AUROC 0.63) and miscalibrated (ECE 0.25). ACI (1.3 % at 45 %) and open-loop conformal (0.0 % at 73.8 %) show a guarantee on the prediction not translating into a useful decision.

B. Do open-loop metrics choose the closed-loop trigger? (RQ2)

Ranking six trigger scores by open-loop AUROC of their scenario peak and by closed-loop unnecessary stops at each certified threshold (Supplementary Section S14), the rankings agree at the top and diverge below it: Kendall's $\tau = 0.60$ on Argoverse 2, 0.47 on nuScenes and 0.73 on Waymo. H1 is refuted only if $\tau \geq 0.8$ and the top score coincides everywhere, so it is not refuted, but the evidence is weak. The confidence score has AUROC 0.45, below chance, yet a threshold on it reaches the miss target at 75.8 % stops by firing indiscriminately.

C. Ablations and latency (RQ3)

In ablations of the geometric score (Supplementary Section S14), a route-following corridor fires in essentially every scenario (84.9 % stops) and confidence or ensemble spread costs 44–47 points of stops. With decisions at 2 Hz, one tick (0.5 s) of delay lifts the alert-level miss rate above the target (5.3 %) and two ticks give 9.1 % (Fig. 3b), and the zero-latency certificate is clearly violated at both (53 % and 100 % corrected). Calibrating the alert under the deployed

latency gives 3.6–4.1 % missed at 34.4 % and 42.5 % stops, with violations of 12.0 and 11.5 % uncorrected (ambiguous) and 1.5 and 0.5 % corrected.

D. Outcome-level certification, with and without latency (RQ3, continued)

The alert certificate leaves 6.5 % harm after the fallback. We therefore certify the residual harm of the executed run at 7.5 % (fixed in Amendment 11: no fallback leaves 15.1 %, always firing 4.1 %, so 5 % is barely reachable, and a later sweep found 7.5 % the lowest target at which every score certifies at zero latency without always firing; Supplementary Section S20) with risk-ordered fixed-sequence testing (Section III-F). At zero latency (calibrate 904, test 2,900) every score certifies and meets the target, the counterfactual-label trigger at 5.4 % realized harm. TTC (10.2 % unnecessary stops) and a supervised time-to-harm trigger (9.6 %) are the cheapest, ahead of the counterfactual-label trigger (13.6 %) and the geometric score (32.8 %). The outcome certificate also holds up under dependence and drift (Supplementary Section S22). Over 200 random resplits its violation frequency is 4.5, 7.5 and 0.0 % for the learned, geometric and TTC scores, and 7.0, 10.0 (at the boundary) and 2.0 % when whole spatial groups are resplit. Without the 3.2 % of scenarios with flagged simulator drift the realized harm is at most 5.6 %, but stops fall, and the learned trigger then stops 6.0 % against 7.2 % for TTC, so this 2 Hz zero-latency ordering depends on those scenarios.

Certifiability map. Table II repeats the outcome certificate on an arm with a forced rollout at every simulator step (10 Hz decisions), so the deployed latency can be applied exactly, and for several targets. The left block gives unnecessary stops at the 7.5 % target, where every cell meets the target (realized harm at most 6.8 %). The right block gives the *frontier*, the lowest of nine targets (5–15 %) that a trigger certifies with at most 50 % stops and a realized harm within the target. At 7.5 %, a TTC threshold is the cheapest trigger through 0.5 s (5.8 % stops without delay, 12.6 % at 0.5 s), and the learned trigger stops 13–19 points more (paired CIs exclude 0), but across 50 resplits it certifies usefully in only 76 % of them at 0.5 s, against 92 % for the geometric and 88 % for the RSS trigger. From 0.8 s (post hoc) TTC certifies only by always firing, while the safe-distance and learned triggers need 30–40 %, and at 1.0 s the learned trigger is the cheapest (9.3 points below the safe-distance trigger, CI [5.7, 12.9]). TTC is not the best at stricter targets, either. From 0.2 to 0.5 s the geometric or learned trigger certifies a target 1–2 points lower than TTC does (5–6 % against 6–7 %), at 36–49 % stops, and at 1.0 s the learned trigger reaches 6 % where TTC needs 12 % (Supplementary Section S21). The practical reading is a map: for moderate targets and delays up to 0.5 s a certified TTC threshold is the cheapest valid trigger, and stricter targets or longer delays need a trigger that anticipates, at a much higher stop cost.

E. Dataset shift (RQ4)

The Argoverse 2 certificate does not transfer. On nuScenes the threshold certified at a 5 % target misses 13.2 % [11.6, 15.0], 2.6 times the target and with an interval that excludes it. The pre-registered hypothesis H3 is **supported** (15.2 % after dropping drift scenarios). Part of this is a floor rather than shift: 5.0 % of nuScenes scenarios already contain harm at the first decision tick, which even a trigger that always fires cannot prevent (0.1 % on Argoverse 2). That floor equals the target, so no threshold can be certified at $\alpha = 0.05$ on nuScenes. With those scenarios excluded (post hoc), the shifted miss rate is 9.2 % [7.8, 10.8], still above the target. These intervals treat scenarios as independent, but the 1,504 nuScenes scenarios come from only 37 scenes. Resampling scenes widens the interval to [6.1, 21.0], which still excludes the target, while a second, disjoint set of 23 scenes gives 9.5 % with scene-level interval [3.9, 16.5], which does not: the direction of the failure replicates but its size is uncertain (Supplementary Section S6). A domain classifier on the covariates separates Argoverse 2 from nuScenes (cross-validated AUROC 0.97) far more easily than from Waymo (0.91), so the certificate failed under the larger measured shift and held under the smaller. Six further natural shifts, the Argoverse 2 cities held out one at a time, add evidence: the certificate met the target in four of six and failed in Miami (6.1 % [4.8, 7.8]) and Washington, DC (5.3 %), the two cities with the highest harm rates. Across these and the two datasets the held-out miss rate tracks the target's harm rate (Spearman 0.90, $n = 8$, post hoc), which gives a deployment check: a target whose harm rate exceeds

TABLE II

OUTCOME-CERTIFIED TRIGGERS ON THE 10 HZ ARM (CALIBRATE 441, TEST 884; 2.2–2.5 % OF SCENARIOS THAT DID NOT REPRODUCE THE STORED REFERENCE RUN ARE DROPPED). LEFT: UNNECESSARY STOPS (%) AT A 7.5 % TARGET. RIGHT: FRONTIER, THE LOWEST TARGET (%) IN $\{5, 6, 7, 7.5, 8, 9, 10, 12, 15\}$ CERTIFIED WITH AT MOST 50 % STOPS AND A REALIZED HARM WITHIN THE TARGET. †: ALWAYS FIRING. GEOMETRIC: PREDICTOR SCORES HELD AT 2 HZ. LEARNED: COUNTERFACTUAL LABELS. BOLD: BEST IN THE ROW. COMPARISONS ARE DESCRIPTIVE AND UNADJUSTED FOR MULTIPLICITY.

	Stops at 7.5 %				Frontier (%)			
Latency	TTC	RSS	Geom.	Learned	TTC	RSS	Geom.	Learned
0 s	**5.8**	20.1	32.9	19.0	**6**	**6**	**6**	**6**
0.2 s	**8.7**	24.7	28.5	25.2	6	7.5	6	**5**
0.3 s	**9.5**	23.9	26.5	28.4	7	7.5	**5**	6
0.5 s	**12.6**	23.6	31.9	25.6	7	7	**6**	**6**
0.8 s	86.7†	**29.9**	86.7†	36.5	8	7.5	9	**6**
1.0 s	86.7†	40.2	86.7†	**30.9**	12	**6**	10	**6**

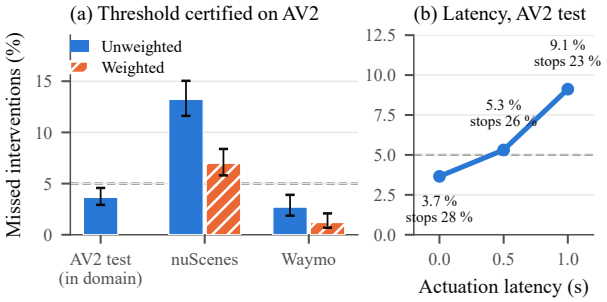


Fig. 3. (a) Miss rate of the Argoverse 2-certified threshold deployed unchanged on nuScenes and Waymo, unweighted and with the density-ratio repair (Section III-D); error bars are 95 % Wilson CIs. (b) Miss rate and unnecessary stops as actuation latency increases (Supplementary Section S14).

TABLE III
A THRESHOLD CERTIFIED ON ARGOVERSE 2 (LTT, $\alpha = 0.05$) DEPLOYED UNCHANGED ON NUSCENES AND WAYMO. "AVOIDABLE" EXCLUDES THE 5.0 % OF NUSCENES SCENARIOS WITH HARM AT THE FIRST DECISION TICK (POST HOC). WEIGHTED: DENSITY-RATIO REPAIR, NO TARGET LABELS.

Setting	Miss (%) [95 % CI]	Unnec. (%)	n
Argoverse 2 test (in domain)	3.7 [2.9, 4.6]	28.5	1,996
nuScenes, unweighted	13.2 [11.6, 15.0]	26.1	1,504
nuScenes, weighted	7.0 [5.8, 8.4]	52.4	1,504
nuScenes, oracle threshold	5.0	77.9	1,504
Avoidable, unweighted	9.2 [7.8, 10.8]	27.1	1,429
Avoidable, weighted ($n_{\text{eff}} = 68$)	2.2 [1.6, 3.1]	54.5	1,429
Waymo, unweighted	2.7 [1.9, 3.9]	45.1	1,000
Waymo, weighted	1.2 [0.7, 2.1]	50.2	1,000
Waymo, oracle threshold	5.0	33.9	1,000

the source's should be audited before the certificate is trusted (Supplementary Section S10).

The outcome certificate under shift. Deployed unchanged on the same eight targets at zero latency, the outcome-certified TTC threshold realizes harm at or below the 7.5 % target on six of eight at 5–14 % unnecessary stops. Its Wilson interval, which treats scenarios as independent, lies below the target for Austin, Dearborn, Palo Alto and Waymo, and contains it for Pittsburgh (6.1 % [4.9, 7.5]) and Washington, DC (6.7 % [5.0, 8.8]). It exceeds the target in Miami (9.2 % [7.8, 10.8]) and, with an interval that contains it, on nuScenes (8.8 % [7.4, 10.3]). The geometric and learned triggers also meet it on six of eight, the geometric score with more stops on every target (Supplementary Section S18).

Density-ratio weighting moves the nuScenes miss rate from 13.2 % to 7.0 %, still above the target, and doubles unnecessary stops. On the avoidable subset it reaches 2.2 % with $n_{\text{eff}} = 68$, almost entirely through conservatism.

F. What the intervention does, and a class-conditional certificate (post hoc)

The certified quantity is the alert. At the certified threshold on Argoverse 2, 3.7 % of scenarios are missed, in a further 1.7 % the fallback fires before harm but harm still occurs, in 9.8 % harm is prevented, in 1.1 % the intervened run is harmful although the unintervened run was not, and 2.6 % suffer an induced collision (Supplementary Section S19). Harm in the intervened run is therefore 6.5 % (15.1 % without a fallback), almost twice the certified miss rate; on the fresh split it is 5.0 % [3.7, 6.6] against 3.4 % missed. Always firing leaves 4.1 % residual harm, 3.1 % of it new, and on nuScenes the same threshold leaves 15.6 %. Induced collisions are other vehicles striking the ego at low to moderate speed (median difference 2.4 m/s), so this cost may partly reflect simulated traffic that does not yield to a stationary vehicle (Supplementary Section S6).

Class-conditional certification. A Luo-style class-conditional certificate reduces to always firing at $\epsilon = 0.05$, and at the marginal-matched $\epsilon = 0.32$ it selects the tuned threshold and misses 66 % of harmful nuScenes scenarios against its 32 % target (Supplementary Section S1).

TABLE IV
ROBUSTNESS AND REPLICATION CHECKS. VIOL.: RESPLIT VIOLATION FREQUENCY OF LTT, UNCORRECTED / CORRECTED; UNNEC.: UNNECESSARY STOPS OF LTT AGAINST THE TUNED CONFIDENCE TRIGGER (T1). RATES IN %.

Check	Setting	Result
Fresh disjoint data	904 / 904 scenarios	Viol. 7.5 / 0.5, miss 3.4
Second AutoBot	minADE 0.854	Viol. 9.0 / 0.0, stops 30.6 vs 68.3
Wayformer	second architecture	Viol. 7.0 / 0.0, stops 41.8 vs 67.8
Non-reactive traffic	log replay	Miss 3.7 [2.2, 6.1], stops 35.4 vs 71.7
Alternative harm	collision only, TTC 0.5 s, 1.5 s	Geometric beats T1/T2 by ≥ 20 points
Held-out cities	6 Argoverse 2 cities	Target met in 4 of 6

G. Second shifted dataset and recalibration

Waymo Open Motion. Table III and Fig. 3a deploy the same Argoverse 2-certified threshold, unchanged, on 1,000 Waymo Open Motion validation scenarios [26] (9.1 s, 1 s of history). The certificate transfers: it misses 2.7 % [1.9, 3.9], with an interval below the 5 % target, so failure under shift is dataset-dependent (and H3-W, which predicted failure, is refuted). Density-ratio weighting lowers the miss rate to 1.2 % at 5 more points of stops.

Waymo recalibration (pre-registered R1–R4). With 40 and 60 labeled Waymo scenes (200 draws; Supplementary Section S3), uncertified tuning violates the target in 69.5 % and 59 % of draws (**R1 holds**), while weighting the Argoverse 2 calibration set with 60 Waymo scenes keeps validity (0 %, **R2 holds**) at 21.3 more points of stops (**R3 holds**); **R4** restates H3-W and is refuted.

H. Robustness checks (post hoc)

Table IV collects the checks that vary the data, predictor, traffic, harm definition or splitting, each registered as a post hoc analysis before it was run. The validity of Learn-then-Test and the advantage of the geometric score over the confidence and ensemble triggers survive all of them, although TTC beats the geometric score under three of the four harm definitions. The cities and spatial groups are where the guarantee weakens (Section V-E).

I. What did not hold

The pre-registered claims below were refuted; the ledger of every claim is in Supplementary Section S17. **H2:** the tuned geometric threshold and CDT stop less often than LTT (paired difference -2.5 points, CI $[-3.2, -1.8]$ against tuned), so certification buys validity, not a better trade-off. **H3-W, cities:** the certificate held on Waymo (Section V-G) and met the target in four of six cities against the 80 % we pre-registered. **Learned triggers:** the counterfactual-label trigger beat neither TTC nor a supervised time-to-harm trigger at zero latency, nor TTC and the safe-distance trigger at 0.3 and 0.5 s with 10 Hz decisions, and TTC did not reach the lowest certifiable target from 0.2 to 0.5 s (Table II). Physics features alone gave the same trigger as all features (24.8 % against 25.5 % stops), so the predictor adds no lead time here. An

Algorithm 1 Certifying the outcome of a fallback trigger

1: Simulate one reference run and one forced rollout per decision tick for every scenario.
2: For a score and latency ℓ, look up the executed outcome $\Phi(X, \tau_\lambda(X)+\ell)$ for every threshold λ.
3: Order the thresholds by outcome risk on a training split, then walk that order on a calibration split with Hoeffding–Bentkus tests at level δ, stopping at the first failure.
4: Deploy the certified threshold with the fewest unnecessary stops. In a new domain, audit on labeled target data when its harm rate exceeds the source's.

apparent advantage at 0.5 s with 2 Hz decisions (26 % against 85 % stops for TTC) came from a certification cliff: the 7.5 % target lies at the risk of a 15 %-stop threshold, so the certified set flips to always firing in some calibration samples. We withdrew it (Supplementary Section S20). **Repairs:** correcting the certificate by a harm prevalence estimated with black-box shift estimation [48] was unreliable, an outcome audit bound covered fewer than 90 % of targets, joint certification of harm and induced collisions succeeded at zero latency but for no score at 0.5 s, and a collision-only harm label did not resolve a 2 % target even with 1,994 calibration scenarios (Supplementary Sections S15, S16, S20 and S21).

VI. Discussion and Scope

How to use the results. Run Algorithm 1 for each candidate, start from a certified TTC threshold, and add an anticipating trigger when the target is strict or the delay reaches 0.8 s.

What we claim, and what we do not. We claim that certification makes a closed-loop trigger keep its stated target, that the executed outcome can be certified with risk-ordered testing, that a certified physics trigger is the one to beat at moderate targets, and that the guarantee is fragile to shift and dependence. We do not claim that a learned trigger is cheaper than TTC at the same target below 0.8 s, or a guarantee under shift.

The harm label and the trigger. Harm is defined by time-to-collision, so a TTC trigger is close to the label and its advantage is partly structural, although it holds at the alert level in three of four harm definitions and, at zero latency, at the outcome level under collision-only harm at a 3 % target (20.7 % against 42.4 % stops; Supplementary Section S21). The simulated ego is a rule-based driver, not a learned planner, so results concern triggering.

Alert versus outcome. The outcome certificate bounds the executed run's harm only as well as the calibration data resemble deployment. The 5.0 % first-tick floor on nuScenes shows that a miss metric on a replayed snippet mixes preventable events with events already under way, so benchmarks should report an avoidable-only variant.

Simulator, perception and dependence. All results use MetaDrive with rule-based (IDM) agents, so whether the ordering of triggers survives learned agent behavior is open, and the induced-collision rate is best read as an upper bound. The predictor receives agent states, which isolates triggering from perception error. Guarantees treat scenarios as exchangeable, which nearby scenarios may violate: our nuScenes samples span only 37 and 23 scenes.

VII. Conclusion

A certificate on a fallback trigger bounds what it was calibrated to bound, and what that is matters. With the fallback executed, a Learn-then-Test certified alert keeps its 5 % target in distribution (3.7 % missed, violations in 9.5 % of resplits against 43.5 % uncertified) but leaves 6.5 % harm after the fallback, and risk-ordered testing certifies the executed outcome directly (5.4 % realized against a 7.5 % target). A map of what can be certified shows that, at a 7.5 % target, a certified time-to-collision threshold is the cheapest trigger up to 0.5 s of latency with decisions at 10 Hz (6–13 % unnecessary stops) and meets its outcome target at zero latency on six of eight shifted test sets, whereas targets 1–2 points stricter need geometric or learned triggers and delays of 0.8 s or more need learned or safe-distance triggers, at 30–50 % stops. The certificate fails on nuScenes and in two of six held-out Argoverse 2 cities, the targets with the highest harm rates, but holds on Waymo. For the setting studied, these findings argue for certifying the outcome at the deployed latency and decision rate, starting from a certified physics trigger, and auditing on data that resemble deployment. The lookup protocol makes each step inexpensive to repeat.

Conflict of Interest

The authors declare no competing financial or personal interests.

References

[1] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. Berlin, Germany: Springer-Verlag, 2005.
[2] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, and L. Lei, "Learn then test: Calibrating predictive algorithms to achieve risk control," *Ann. Appl. Statist.*, vol. 19, no. 2, pp. 1641–1662, 2025.
[3] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster, "Conformal risk control," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024.
[4] L. Lindemann, M. Cleaveland, G. Shim, and G. J. Pappas, "Safe planning in dynamic environments using conformal prediction," *IEEE Robot. Autom. Lett.*, vol. 8, no. 8, pp. 5116–5123, Aug. 2023.
[5] A. Dixit, L. Lindemann, S. X. Wei, M. Cleaveland, G. J. Pappas, and J. W. Burdick, "Adaptive conformal prediction for motion planning among dynamic agents," in *Proc. 5th Annu. Conf. Learn. Dyn. Control (L4DC)*, PMLR vol. 211, 2023, pp. 300–314.
[6] B. Wilson *et al.*, "Argoverse 2: Next generation datasets for self-driving perception and forecasting," in *Proc. 35th Conf. Neural Inf. Process. Syst. (NeurIPS), Track Datasets and Benchmarks*, 2021.
[7] H. Caesar *et al.*, "nuScenes: A multimodal dataset for autonomous driving," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 11621–11631.
[8] M. Treiber, A. Hennecke, and D. Helbing, "Congested traffic states in empirical observations and microscopic simulations," *Phys. Rev. E*, vol. 62, no. 2, pp. 1805–1824, 2000.
[9] A. N. Angelopoulos, R. Foygel Barber, and S. Bates, "Theoretical foundations of conformal prediction," Cambridge Univ. Press (to appear), *arXiv preprint arXiv:2411.11824*, 2024.
[10] L. Lindemann, Y. Zhao, X. Yu, G. J. Pappas, and J. V. Deshmukh, "Formal verification and control with conformal prediction," *arXiv preprint arXiv:2409.00536*, 2024.
[11] Y. Shu, J. Xu, L. Kong, F. Zhang, G. Yin, and C. Sun, "Scenario-aware uncertainty quantification for trajectory prediction with statistical guarantees," *arXiv preprint arXiv:2512.05682*, 2025.
[12] H. Wang and C. Ning, "Conformal prediction in the loop: A feedback-based uncertainty model for trajectory optimization," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2025.
[13] P. Y. Chang and Q. Ahmed, "Barrier function conformal safety clearance certification with CVaR for driving trajectory selection," *arXiv preprint arXiv:2608.26533*, 2026.

[14] C. Guo *et al.*, "CoCoNav: Conformal control for safe robot navigation in crowds," *arXiv preprint arXiv:2608.07751*, 2026.

[15] A. N. Angelopoulos, "Conformal risk control for non-monotonic losses," *Phil. Trans. R. Soc. A*, 2026, doi: 10.1098/rsta.2025.0068.

[16] S. Joshi, Y. Sun, H. Hassani, and E. Dobriban, "MultiRisk: Multiple risk control via iterative score thresholding," *arXiv preprint arXiv:2512.24587*, 2025.

[17] J. Gonzales, K. Mizuta, K. Leung, and L. J. Ratliff, "Safe probabilistic planning for human-robot interaction using conformal risk control," *arXiv preprint arXiv:2603.10392*, 2026.

[18] S. Joshi, T. Wang, H. Hassani, and E. Dobriban, "Risk-controlled post-processing of decision policies," *arXiv preprint arXiv:2605.06479*, 2026.

[19] H. Atoui and P. Resende, "Minimum risk maneuver fallback strategy for autonomous vehicles: Design and experimental validation," in *Proc. IEEE Intell. Veh. Symp. (IV)*, 2025, pp. 1979–1984, doi: 10.1109/IV64158.2025.11097617.

[20] X. Zhao, R. Aghazadeh-Chakherlou, C.-H. Cheng, P. Popov, and L. Strigini, "On the need for a statistical foundation in scenario-based testing of autonomous vehicles," in *Proc. IEEE Int. Conf. Intell. Transp. Syst. (ITSC)*, 2025.

[21] T. Puphal, V. Ramtekkar, and K. Nishimiya, "Risk-based filtering of valuable driving situations in the Waymo Open Motion Dataset," *arXiv preprint arXiv:2506.23433*, 2025.

[22] J. Shi *et al.*, "Motion forecasting for autonomous vehicles: A survey," *arXiv preprint arXiv:2502.08664*, 2025.

[23] K. Rahaman, J. V. Deshmukh, A. R. Hota, and L. Lindemann, "When environments shift: Safe planning with generative priors and robust conformal prediction," in *Proc. 8th Annu. Conf. Learn. Dyn. Control (L4DC)*, 2026.

[24] A. Doula, M. Mühlhäuser, and A. S. Guinea, "SafePath: Conformal prediction for safe LLM-based autonomous navigation," *arXiv preprint arXiv:2505.09427*, 2025.

[25] A. Farid, S. Veer, B. Ivanovic, K. Leung, and M. Pavone, "Task-relevant failure detection for trajectory predictors in autonomous vehicles," in *Proc. 6th Conf. Robot Learn. (CoRL)*, Auckland, New Zealand, 2022.

[26] S. Ettinger *et al.*, "Large scale interactive motion forecasting for autonomous driving: The Waymo Open Motion Dataset," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, arXiv:2104.10133.

[27] R. Luo, S. Zhao, J. Kuck, B. Ivanovic, S. Savarese, E. Schmerling, and M. Pavone, "Sample-efficient safety assurances using conformal prediction," *Int. J. Robot. Res.*, vol. 43, no. 9, pp. 1409–1424, 2024.

[28] M. P. Chapman, R. Bonalli, K. M. Smith, I. Yang, M. Pavone, and C. J. Tomlin, "Risk-sensitive safety analysis using conditional value-at-risk," *IEEE Trans. Autom. Control*, vol. 67, no. 12, pp. 6521–6536, Dec. 2022.

[29] Q. Li, Z. Peng, L. Feng, Z. Liu, C. Duan, W. Mo, and B. Zhou, "ScenarioNet: Open-source platform for large-scale traffic scenario simulation and modeling," in *Proc. 37th Conf. Neural Inf. Process. Syst. (NeurIPS), Track Datasets and Benchmarks*, 2023.

[30] N. Karnchanachari *et al.*, "Towards learning-based planning: The nuPlan benchmark for real-world autonomous driving," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2024.

[31] J. Lekeufack, A. N. Angelopoulos, A. Bajcsy, M. I. Jordan, and J. Malik, "Conformal decision theory: Safe autonomous decisions from imperfect predictions," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2024, pp. 11668–11675.

[32] I. Gibbs and E. Candès, "Adaptive conformal inference under distribution shift," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 34, 2021, pp. 1660–1672.

[33] R. J. Tibshirani, R. Foygel Barber, E. J. Candès, and A. Ramdas, "Conformal prediction under covariate shift," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 32, 2019.

[34] Y. Gal and Z. Ghahramani, "Dropout as a Bayesian approximation: Representing model uncertainty in deep learning," in *Proc. 33rd Int. Conf. Mach. Learn. (ICML)*, 2016, pp. 1050–1059.

[35] R. Girgis, F. Golemo, F. Codevilla, M. Weiss, J. A. D'Souza, S. E. Kahou, F. Heide, and C. Pal, "Latent variable sequential set transformers for joint multi-agent motion prediction," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022.

[36] L. Feng, M. Bahari, K. M. Ben Amor, É. Zablocki, M. Cord, and A. Alahi, "UniTraj: A unified framework for scalable vehicle trajectory prediction," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2024.

[37] United Nations Economic Commission for Europe, "UN Regulation No. 157 — Uniform provisions concerning the approval of vehicles with regard to Automated Lane Keeping Systems," Rev. 1, 2025.

[38] R. Sinha, E. Schmerling, and M. Pavone, "Closing the loop on runtime monitors with fallback-safe MPC," in *Proc. IEEE Conf. Decis. Control (CDC)*, 2023.

[39] S. Sohrabi *et al.*, "Quantifying the automated vehicle safety performance: A scoping review of the literature, evaluation of methods, and directions for future research," *Accid. Anal. Prev.*, vol. 152, Art. no. 106003, 2021.

[40] D. Xie *et al.*, "A survey on scenario-based test and evaluation methods for autonomous vehicles," *Proc. Inst. Mech. Eng. D, J. Automobile Eng.*, 2025.

[41] S. Shalev-Shwartz, S. Shammah, and A. Shashua, "On a formal model of safe and scalable self-driving cars," *arXiv preprint arXiv:1708.06374*, 2017, doi: 10.48550/arXiv.1708.06374.

[42] International Organization for Standardization, "ISO 21448:2022, Road vehicles—Safety of the intended functionality (SOTIF)," ISO, 2022.

[43] UL Standards & Engagement, "UL 4600, Standard for safety for the evaluation of autonomous products," 3rd ed., Mar. 2023.

[44] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017, pp. 4878–4887.

[45] K. Miller, C. K. Zeitler, W. Shen, K. Hobbs, S. Mitra, J. Schierman, and M. Viswanathan, "Searching for optimal runtime assurance via reachability and reinforcement learning," *arXiv preprint arXiv:2310.04288*, 2023.

[46] A. O. Feldman, D. I. Harp, J. Duncan, and M. Schwager, "Conformal safety monitoring for flight testing: A case study in data-driven safety learning," *arXiv preprint arXiv:2511.20811*, 2025.

[47] N. Nayakanti, R. Al-Rfou, A. Zhou, K. Goel, K. S. Refaat, and B. Sapp, "Wayformer: Motion forecasting via simple and efficient attention networks," *arXiv preprint arXiv:2207.05844*, 2022.

[48] Z. Lipton, Y.-X. Wang, and A. Smola, "Detecting and correcting for label shift with black box predictors," in *Proc. 35th Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 80, 2018, pp. 3122–3130.

[49] L. Sha, "Using simplicity to control complexity," *IEEE Softw.*, vol. 18, no. 4, pp. 20–28, Jul./Aug. 2001, doi: 10.1109/MS.2001.936213.

Hassan Waqas Saleem received the M.S. degree in electrical engineering, with a specialization in artificial intelligence and autonomous systems, from the National University of Sciences and Technology (NUST), Pakistan, the B.Sc. degree in electrical engineering from the University of Engineering and Technology, Taxila, Pakistan, and the B.Tech. degree in electrical technology from Bahauddin Zakariya University, Multan, Pakistan. He is currently a Technical Lead with a public-sector organization in Pakistan. His research interests include artificial intelligence for autonomous systems, uncertainty quantification and safety assurance for automated driving, graph neural networks, knowledge graph completion, large language models, robotics, and simultaneous localization and mapping.

Rahat Zeeshan Saleem received the M.S. degree in supply chain management from the NUST Business School, National University of Sciences and Technology (NUST), Islamabad, Pakistan, in 2023, and the M.B.A. degree from Bahauddin Zakariya University, Multan, Pakistan, in 2021, where he was awarded a gold medal. He is currently a Project Lead with a private-sector organization in Pakistan. His research interests include artificial intelligence and data-driven decision support.